



Review Test Submission: Assignment Week 1

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Course EBN 122 S2 2017

Test Assignment Week 1

Started 7/23/17 3:13 PM

Submitted 7/23/17 4:14 PM

Due Date 7/24/17 11:59 PM

Status Completed

Attempt Score 14.5 out of 16 points

Time 1 hour, 1 minute

Elapsed

Instructions Read over the document *ClickUP Assignments Instructions*.

Save the answer to each question as you complete it.

At the end of the assignment click Save and Submit.

Give your final correct answer 3 significant digits.

Make sure you use the FULL STOP for your decimal place marker.

You must indicate the units, unless otherwise specified:

- There must be NO space between the answer
- There must be between 1 and 3 digits to the left of the decimal point. *This means that you need enter 1kV rather than 1000V and 100mV rather than 0.1V*
- Write out 'ohm' to indicate Ω
- Use 'u' to indicate micro.
- Only use pico(**p**), nano(**n**), micro(**u**), milli(**m**), kilo(**k**), mega(**M**), giga(**G**) and tera(**T**); do not use centi, deci and deca

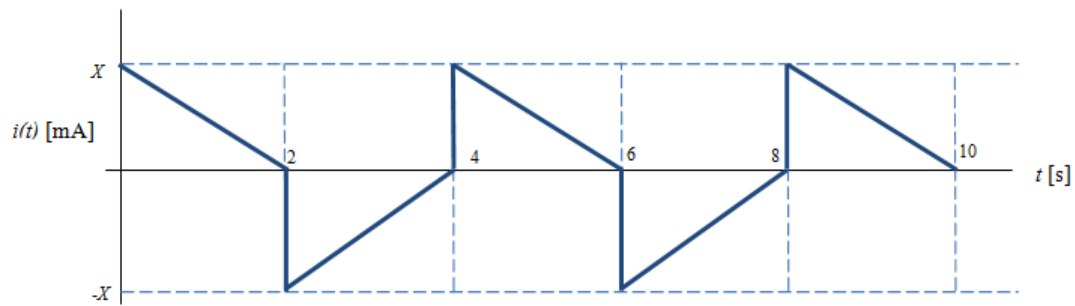
Results All Answers, Submitted Answers, Feedback, Incorrectly Answered Questions
Displayed

Question 1

2 out of 2 points



The graph shows the current through an element over time. Determine the charge that passes through the element between 2 and 8 seconds for a current of 25 mA.



Selected Answer: -25mC

Question 2

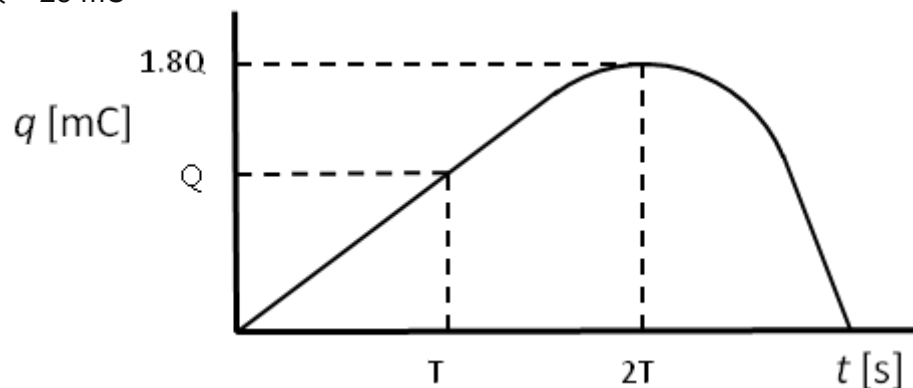
2 out of 2 points



The charge flowing in a conductor is shown below. Determine the current at $t = 2T$ s.
(The curve reaches its maximum value at $t = 2T$ s.)

$T = 0.6$ s

$Q = 25$ mC



Selected Answer: 0A

Question 3

2 out of 2 points



To move -65 nC in 9.1 micro seconds through an element requires 541 mJ.
Determine the current through the element, assuming a constant current.

Selected Answer: -7.14 mA

Question 4

2 out of 2 points



A rechargeable flashlight battery is capable of delivering 3 mA for about 10 hours. If the terminal voltage is 1 V, how much energy can the battery deliver?

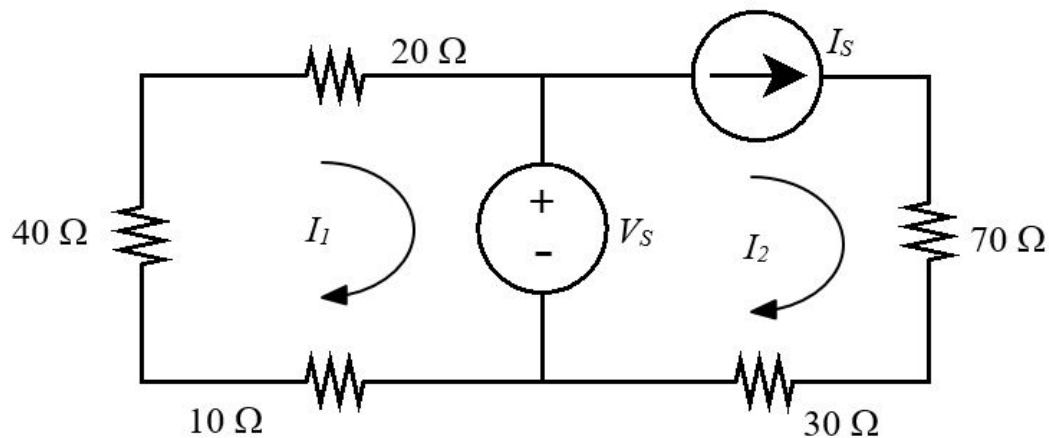
Selected Answer: 108 J

Question 5

2 out of 2 points



How many nodes (n) [n], independent loops (l) [l], and branches (b) [b] are there in the following circuit?



Specified Answer for: n 6

Specified Answer for: l 2

Specified Answer for: b 7

Question 6

0.5 out of 2 points



The voltage across a $33\ \Omega$ resistor increases linearly from 0 mV to 21 mV between $t = 0\text{ s}$ and $t = 20\text{ s}$.

Calculate the value of the charge flowing through the resistor between $t = 5\text{ s}$ and $t = 13\text{ s}$.

Selected Answer: 2.93mC

Question 7

2 out of 2 points



An electric heater is specified as 224 V; 2.5 kW. What is the current flowing through the heater?

Selected Answer: 11.2A

Question 8

0 out of 0 points



The voltage across an element is $v(t) = \frac{di}{dt}$ V and the current through the element is

$i(t) = 2e^{-4t}$ mA. Determine the power $p(t)$ (in μW) (associated with the element as a function of time. The current flows from the negative to the positive potential through the element. (In other words, the current is entering the negative terminal and exiting the positive terminal.)

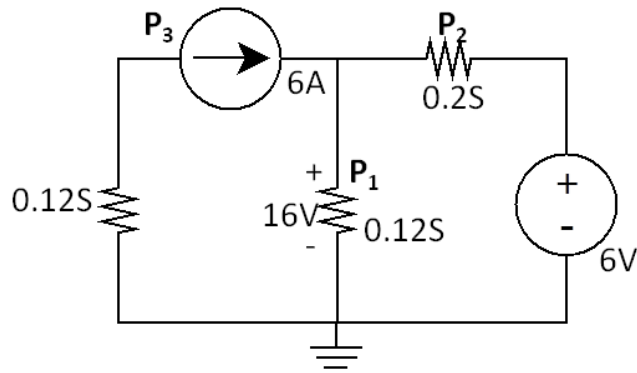
The answer is in the form Ae^{Bt} with $A=[\mathbf{A}]$ and $B=[\mathbf{B}]$

Specified Answer for: A -16

Specified Answer for: B -8

Question 9

0 out of 0 points



Calculate the power P_2 Dissipated in the resistor.

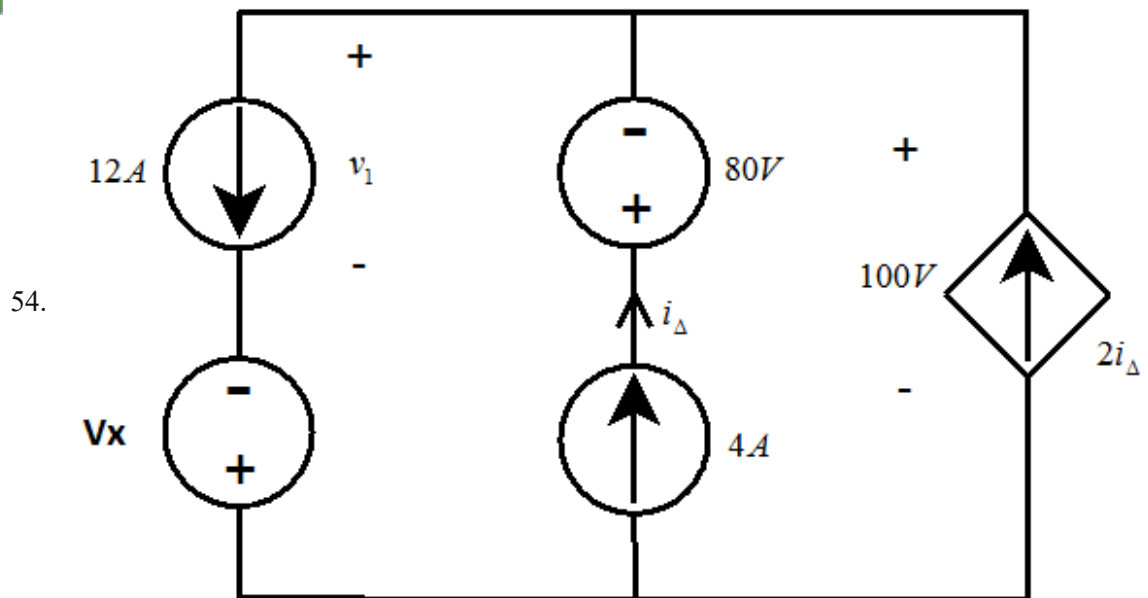
Selected Answer: 83.2W

Question 10

2 out of 2 points



Calculate the power associated with the voltage source on the left most side of the circuit if $V_x =$



Selected Answer: -648W

Sunday, July 23, 2017 4:14:57 PM SAST

← OK